Year 1 AS Pure Maths

Chapter 2

Quadratics

Practice Booklet

Contents

2.1 Solving quadratic equations ..... book p. 7
2.2 Completing the square ..... book p. 8
2.3 Functions ..... book p. 8
2.4 Quadratic graphs ..... book p. 9
2.5 The discriminant ..... book p. 10
2.6 Modelling with quadratics ..... book p. 11
Problem solving: Set A ..... book p. 14
Problem solving: Set B ..... book p. 15
Selected exam question bank questions .....
Answers .....

2.1 Solving quadratic equations

Practice Book, printed page 7.

1. Solve the following equations using factorisation:

a. $x^2 + 7x + 10 = 0$

b. $x^2 - 5x - 24 = 0$

c. $x^2 + 6x = 0$

Hint

The factorised quadratic equation $(x - a)(x - b) = 0$ will have solutions $x = a$ and $x = b$.

2. Solve the following equations using factorisation:

a. $7x^2 = 21x$

b. $4x^2 = 49$

c. $4x^2 - 20x + 25 = 0$

d. $2x^2 - 5x + 2 = 0$

Hint

Write the equation in the form $ax^2 + bx + c = 0$ before factorising. Solutions to quadratic equations do not have to be integers.

3. Solve the following equations:

a. $(x - 4)^2 = 0$

b. $x(2x - 1) = 21$

c. $4x^2 + 4x + 24 = 2x^2 - 10x$

Hint

A quadratic equation may have only one solution, called a repeated root: If $(x - a)^2 = 0$, then $x = a$.

4. Solve the following equations using the quadratic formula. Give your answers exactly, leaving them in surd form.

a. $x^2 + 8x + 6 = 0$

b. $x^2 + 4x = 1$

c. $2x^2 - 12x + 15 = 0$

Hint

The solutions to $ax^2 + bx + c = 0$ are given by $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

5. Solve the following equations using a suitable method. Where necessary, give your answers to 3 significant figures.

a. $10x - x^2 - 9 = 0$

b. $64x^2 = 100$

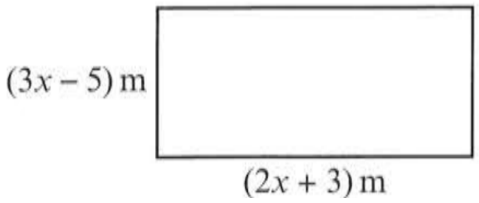
c. $3x^2 + 10x - 25 = 0$

d. $12x = 3x^2 + 5$

e. $x = \sqrt{5x}$

f. $2x^2 + 6x + 1 = 0$

6. E The length of a rectangular carpet is $(2x+3)$ metres and its width is $(3x-5)$ metres. The carpet has an area of 20 m^2.



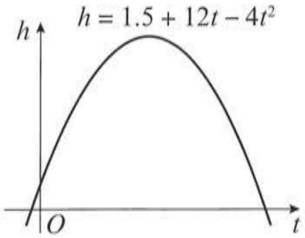
a. Show that $6x^2 - x - 35 = 0$. (2 marks)

b. Hence find the length and width of the carpet, in metres. (2 marks)

7. E Solve the equation $x^2 + 4x + 1 = 0$. Write your answer in the form $a \pm \sqrt{b}$, where a and b are integers to be found. (2 marks)

8. E Solve the equation $0.1x^2 + 1.6x = 0.8$. Give your answers to 3 significant figures. (2 marks)

9. E/P The height, h metres, of a ball at time t seconds after it is thrown up in the air can be modelled by the equation $h = 1.5 + 12t - 4t^2$.



a. Find how many seconds it takes for the ball to hit the ground again. Give your answer correct to 3 significant figures. (2 marks)

b. Find the times when the height of the ball is 9.5 m. (2 marks)

2.2 Completing the square

Practice Book, printed page 8.

1. Complete the square for these expressions:

a. $x^2 + 2x$

b. $x^2 - 8x$

c. $x^2 - 12x$

Hint

$$x^2 + bx \equiv \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$$

2. Complete the square for these expressions:

a. $x^2 - 8x + 12$

b. $x^2 - x - 12$

c. $x^2 - 3x - 4$

Hint

When a quadratic expression is in completed square form, x only appears once. The constants in your answer do not have to be integers.

3. Write each of these expressions in the form $a(x+b)^2 + c$, where a, b and c are constants to be found.

a. $3x^2 - 12x + 17$

b. $5x^2 - 10x + 12$

c. $3x^2 - 7x + 2$

Hint

Start by factorising the first two terms of each expression.

4. Solve these quadratic equations by completing the square. Leave your answers in surd form.

a. $x^2 - 2x - 10 = 0$

b. $5x^2 + 12x + 6 = 0$

c. $3x^2 - 7x - 2 = 0$

Hint

Write the left-hand side of each equation in completed square form. Then use inverse operations to make x the subject of the equation. Remember to include $\pm$ when you take square roots of both sides of the equation.

5. E/P $f(x) = x^2 - 7x - 2$, $x \in \mathbb{R}$

a. Express $f(x)$ in the form $(x-a)^2 - b$, where a and b are constants. (2 marks)

b. Hence write down the minimum value of $f(x)$. (1 mark)

6. E $h(x) = 4 - 2x - 3x^2$, $x \in \mathbb{R}$

a. Express $h(x)$ in the form $a(x+b)^2 + c$, where a, b and c are constants. (2 marks)

b. Hence, or otherwise, find the exact solutions to $h(x) = 0$. (2 marks)

7. E $6x - 2 - x^2 \equiv q - (x+p)^2$, where p and q are integers.

a. Find the value of p and the value of q. (3 marks)

b. Hence, or otherwise, solve the equation $6x - 2 - x^2 = 0$. (2 marks)

2.3 Functions

Practice Book, printed page 8–9.

1. Using the functions $f(x) = x^2 - 5x + 6$ and $g(x) = |\sqrt{x+3}|$, find the values of:

a. $f(3)$

b. $g(22)$

c. $f(-1.5)$

d. $f(4) + g(6)$

Hint

To find the value of $f(a)$, substitute a for x in the expression for $f(x)$.

2. The functions r and s are given by $r(x) = x^2 + 2x - 3$ and $s(x) = 2x + 1$, $x \in \mathbb{R}$. Find the two values of x for which $r(x) = s(x)$.

Hint

You can check your answer by evaluating $r(x)$ and $s(x)$ for each of your values of x.

3. Find all roots of the following functions. Give any non-integer roots in exact form.

a. $f(x) = (x + 2)(x + 7)$

b. $f(x) = 81 - x^2$

c. $f(x) = x^6 + 7x^3 + 12$

d. $f(x) = x^3 - 4x^2 - 21x$

Hint

The roots of a function are the values of x for which $f(x) = 0$.

4. The function $h(x)$ is defined as $h(x) = x^2 + 6x + 2$, $x \in \mathbb{R}$.

a. Write $h(x)$ in the form $(x + p)^2 + q$.

b. Hence, or otherwise, find the roots of $h(x)$, leaving your answers in surd form.

c. Write down the minimum value of $f(x)$ and state the value of x for which it occurs.

Hint

The expression $(x + p)^2$, where x is a real number and p is a constant, must always be greater than or equal to 0.

5. E/P The function f is defined as $f(x) = x^2 + 6x + 13$, $x \in \mathbb{R}$.

a. Write $f(x)$ in the form $(x + p)^2 + q$, where p and q are constants to be found. (2 marks)

b. Hence, or otherwise, explain why $f(x) > 0$ for all values of x and find the minimum value of $f(x)$. (2 marks)

6. E/P Find all real roots of the following functions:

a. $f(x) = x^8 + 6x^4 - 7$ (2 marks)

b. $f(x) = 6x^{10} - 5x^5 + 1$ (2 marks)

c. $f(x) = x^{\frac{1}{4}} - 3x^{\frac{1}{2}} + 2$ (2 marks)

7. E/P The function f is defined as $f(x) = 2^{2x} - 6(2^x) + 8$, $x \in \mathbb{R}$

a. Write $f(x)$ in the form $(2^x - a)(2^x - b)$, where a and b are real constants. (2 marks)

b. Hence find the two roots of $f(x)$. (2 marks)

2.4 Quadratic graphs

Practice Book, printed page 9–10.

1. Sketch graphs of each of the following equations, showing the coordinates of the points where the graph crosses the coordinate axes.

a. $y = x^2 + 11x + 18$

b. $y = 4x^2 - 16$

c. $y = -6x^2 + 2x$

Hint

Factorise each equation to find the points where $y = 0$. These are the values of x at the points where the graph crosses the x-axis. To find the y-intercept, substitute $x = 0$ into the equation.

2. Find the coordinates of the turning point on each of the graphs:

a. $y = (x - 1)^2 + 9$

b. $y = x^2 + x - 6$

c. $y = -x^2 - 13x - 42$

Hint

You can find coordinates of the turning point on a quadratic curve by completing the square. The curve with equation $y = (x - a)^2 + b$ will have a turning point at (a, b).

3. Sketch the graphs of the following equations. For each graph, indicate where the graph crosses the coordinate axes, and write down the coordinates of the turning point and the equation of the line of symmetry.

a. $y = x^2 - 6x + 20$

b. $y = -2x^2 - 5x - 2$

c. $4x^2 - y = 4x + 3$

Hint

A quadratic graph has a vertical line of symmetry that passes through its turning point.

4. Sketch the graphs of the following equations. For each graph, indicate where the graph crosses the coordinate axes, leaving your answer in surd form. Write down the coordinates of the turning point and the equation of the line of symmetry.

a. $y = x^2 + 7x + 5$

b. $y = -5x^2 - 12x - 3$

c. $y = 2x^2 + 7x + 4$

Hint

After drawing your axes, write in all the required coordinate points and then draw a smooth curve through these points.

5. E The expression $8x - 7 - x^2$ can be written in the form $q - (x - p)^2$, where p and q are integers.

a. Find the value of p and the value of q. (3 marks)

b. Sketch the curve with equation $y = 8x - 7 - x^2$, showing clearly the coordinates of any points where the curve crosses the coordinate axes. (3 marks)

6. E $f(x) = x^2 + 6x + 4$, $x \in \mathbb{R}$

a. Express $f(x)$ in the form $(x + a)^2 + b$, where a and b are constants. (2 marks)

The curve C with equation $y = f(x)$ crosses the y-axis at point P and has a minimum point at the point Q.

b. Sketch the graph of C, showing the coordinates of points P and Q. (3 marks)

c. Explain why the equation $f(x) = -6$ has no real solutions. (1 mark)

7. E/P $p(x) = 3 - 2x$, $q(x) = x^2 - 9x - 10$, $x \in \mathbb{R}$

a. Solve the equation $q(x) = 0$. (2 marks)

b. Sketch the graphs of $y = p(x)$ and $y = q(x)$ on the same set of axes. Label all points where the curves intersect the coordinate axes. (4 marks)

8. E/P The graph of $y = ax^2 + bx + c$ has a minimum at $(2, -5)$ and passes through $(3, 0)$. Find the values of a, b and c. (4 marks)

2.5 The discriminant

Practice Book, printed page 10–11.

1. State the condition for which the function $f(x) = ax^2 + bx + c$ has:

a. 2 distinct, real roots

b. 1 repeated root

c. no real roots

Hint

The discriminant of the quadratic function $f(x) = ax^2 + bx + c$ is $b^2 - 4ac$.

2. For each of these functions:
i calculate the value of the discriminant
ii write down the number of real roots of the function.

a. $f(x) = -2x^2 - 11x - 12$
b. $f(x) = x^2 + 6x + 9$
c. $f(x) = 3x^2 - 12x + 18$

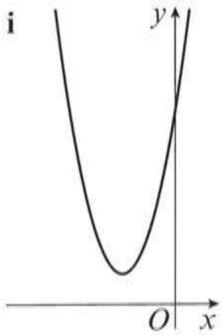
Hint

Be careful with negative signs. In part **a**, $a = -2$, $b = -11$ and $c = -12$.

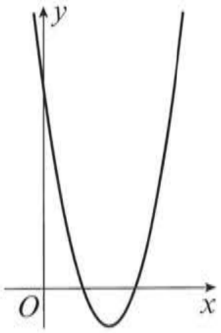
3. Calculate the value of the discriminant for each of these functions and match them to the sketch graphs.

a. $f(x) = x^2 - 10x + 21$
b. $f(x) = -x^2 - 10x - 25$
c. $f(x) = x^2 + 8x + 19$

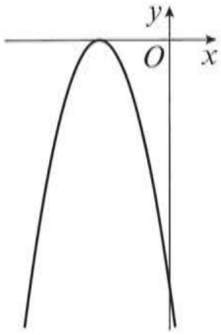
i



ii



iii



Hint

A quadratic graph with one repeated real root will have its turning point on the x-axis.

4. E/P $f(x) = x^2 + kx + 25$, $x \in \mathbb{R}$

a. Find the discriminant of $f(x)$ in terms of k. (2 marks)
b. Given that the equation $f(x) = 0$ has one repeated root, find the possible values of k. (2 marks)

5. E/P $f(x) = x^2 + (k+3)x + k$, where k is a real constant and $x \in \mathbb{R}$.

a. Find the discriminant of $f(x)$ in terms of k. (2 marks)
b. Show that the discriminant of $f(x)$ can be expressed in the form $(k+a)^2 + b$, where a and b are constants to be found. (2 marks)
c. Show that, for all values of k, the equation $f(x) = 0$ has distinct real roots. (2 marks)

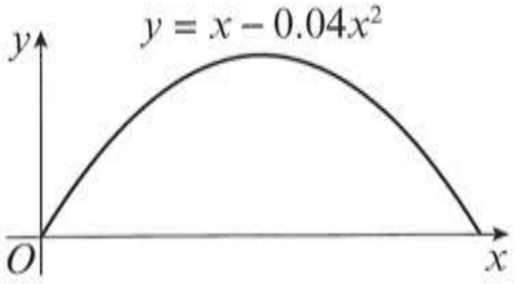
6. E/P The equation $kx^2 + 3x - 5 = 0$, where k is a constant, has two distinct real roots. Find the range of possible values of k. (3 marks)

7. E/P Find the range of values of p for which the equation $2x^2 + 7x + p = 0$ has no real solutions. (3 marks)

2.6 Modelling with quadratics

Practice Book, printed page 11–13.

1. E A ball is kicked through the air from a level playing field. The path of the ball can be modelled by the equation $y = x - 0.04x^2$, where x is the horizontal distance (in metres) and y is the vertical height (in metres) from the point where it was kicked.



a. Find the horizontal distance travelled by the ball from the point where it was kicked to the point where it lands on the playing field again. (2 marks)

b. Show that the equation for the path of the ball can be written in the form $y = a(x - b)^2 + c$, where a, b and c are constants to be found. (3 marks)

c. Using your answer to part **b**, or otherwise, find the maximum height the ball reaches above the playing field, and the horizontal distance at which this maximum height is reached. (2 marks)

Hint

The maximum height of the ball will occur at the turning point of the path.

2. E A company makes a particular type of phone case. The annual profit made by the company is modelled by the equation

$$P = 73.5x - 5.25x^2 - 107.5$$

where P is the annual profit, measured in thousands of pounds, and x is the selling price of the phone case, in pounds. The company wishes to maximise its annual profit.

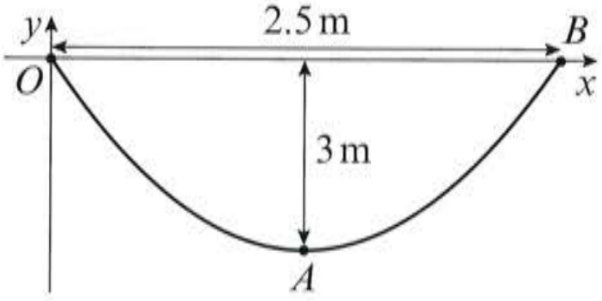
a. Show that this model can be written in the form $P = A - 5.25(x - B)^2$, where A and B are constants to be determined. (2 marks)

b. Hence, or otherwise, state, according to the model:
i the maximum possible annual profit
ii the selling price of the phone case that maximises the annual profit. (2 marks)

Hint

For part **a** you need to complete the square.

3. E/P The diagram shows the cross-section of a canal. O and B are points on the surface of the water, with $OB = 2.5$ m. At point A the canal is 3 m deep. The cross-section is modelled as a quadratic curve OAB. x and y are the horizontal and vertical distances from O in metres.



a. **i** Write down the coordinates of point A.
ii Hence find an equation for the curve OAB. (3 marks)

The cross-section of a narrowboat is modelled as a rectangle of width 2.1 m, with constant depth below water of 1 m.

b. Determine whether or not it is possible for the narrowboat to pass through the canal. (4 marks)

4. E/P A car manufacturing company uses a model to determine how the number of workers employed, w, affects the number of cars produced per day, q. The model suggests $q = 0.4w - 0.00008w^2$.

a. Suggest a reason why this model does not contain a constant term. (1 mark)

On a particular day, 1500 workers are employed.

b. Find the number of cars produced on this day. (1 mark)

c. According to the model, what is the maximum number of cars that can be produced and how many workers are needed for this? (3 marks)

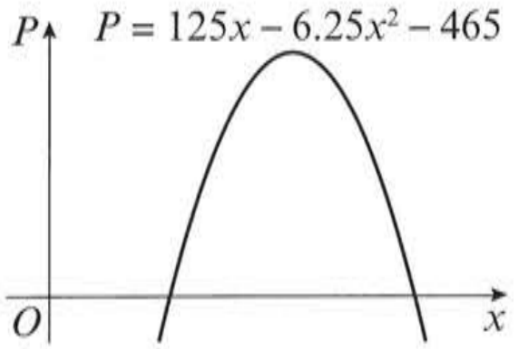
5. E/P A ball is thrown from the top of a cliff. The height, h metres, of the ball above ground level after t seconds is modelled by the function $h(t) = 120 + 12.25t - 4.9t^2$.

a. Give a physical interpretation of the meaning of the constant term 120 in the model. (1 mark)

b. Write $h(t)$ in the form $A - B(t - C)^2$, where A, B and C are constants to be found. (3 marks)

c. Using your answer to part **b**, or otherwise, find, with justification:
i the time between the instant the ball is thrown and the instant it reaches ground level
ii the maximum height of the ball above the ground and the time at which this maximum height is reached. (5 marks)

6. E/P A company makes a particular type of T-shirt. The annual profit made by the company is modelled by the equation $P = 125x - 6.25x^2 - 465$, where P is the profit measured in thousands of pounds and x is the selling price of the T-shirt in pounds. A sketch of P against x is shown in the diagram.



Ellie claims that this model is not valid for a selling price of £4.00, because the value of P is negative.

a. Suggest a reason why Ellie may be incorrect. (1 mark)

b. Write the model in the form $P = c - a(x - b)^2$, where a, b and c are constants to be found. (3 marks)

c. Given that the company made an annual profit of more than £120 000 find, according to the model, the least possible selling price for the T-shirt. (3 marks)

The company wishes to maximise its annual profit.

d. State, according to the model:

i the maximum possible annual profit

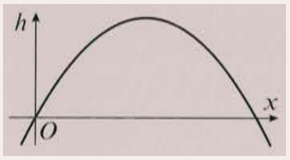
ii the selling price of the T-shirt that maximises the annual profit. (2 marks)

7. P In microeconomics, the market price and quantity of an item are found by relating two functions.
Inverse supply expresses supply price £P_s in terms of the quantity supplied, q thousands.
Inverse demand expresses demand price £P_d in terms of the quantity demanded, q thousands.
Market equilibrium occurs when $P_s = P_d$.
In this model:
Inverse supply is $P_s = 8q + 8$
Inverse demand is $P_d = q^2 - 14q + 48 \qquad (0 \leqslant q \leqslant 6)$
Find the market equilibrium quantity and price given by this model.

Problem solving: Set A

Bronze

The sketch shows the path of a stone that is kicked through the air from level ground. The path of the stone can be modelled by the function $h(x) = 2x - x^2$, where x metres is the horizontal distance the stone travels from the place where it was kicked, and h metres is the vertical height of the stone above ground level.



a. Write $h(x)$ in the form $h(x) = A - (x - B)^2$, where A and B are constants to be found. (3 marks)

b. Using your answer to part **a**, or otherwise, solve $h(x) = 0$ and find the horizontal distance the stone has travelled when it lands on the ground. (3 marks)

Silver

A stone is thrown from the top of a cliff.
The path of the stone can be modelled by the function $h(x) = 114 + 10.4x - 5.2x^2$, where x metres is the horizontal distance the stone travels and h metres is the vertical height of the stone above ground level.

a. Give a physical interpretation of the meaning of the constant term 114 in the model. (1 mark)

b. **i** Show that $h(x)$ can be rearranged to give $h(x) = 114 - 5.2(x^2 - 2x)$.
ii Hence, or otherwise, write $h(x)$ in the form $h(x) = A - 5.2(x - B)^2$, where A and B are constants to be found. (3 marks)

c. Using your answer to part **b ii**, or otherwise, find, with justification:
i the horizontal distance the stone has travelled when it lands on the ground
ii the maximum height of the stone above the ground and the horizontal distance at which this maximum height is reached. (5 marks)

Gold

A stone is thrown from the top of a cliff.
The path of the stone can be modelled by the function $h(x) = 125 + 12.75x - 4.5x^2$, where x metres is the horizontal distance the stone travels and h metres is the vertical height of the stone above ground level.

a. Give a physical interpretation of the meaning of the constant term 125 in the model. (1 mark)

b. Find, with justification, the maximum height of the stone above the ground. (3 marks)

c. If measured in a straight line, what is the distance from the point where the stone is thrown to the point where it lands on the ground? (2 marks)

Problem solving: Set B

Bronze

The equation $2kx^2 + 4x + k = 0$, where k is a constant, has one repeated root.

a. Show that $16 - 8k^2 = 0$. (2 marks)

b. Hence, find two possible values of k. (1 mark)

Silver

The equation $3x^2 + px + 2p = 0$, where p is a non-zero constant, has equal roots.
Find the value of p. (3 marks)

Gold

The equation $\dfrac{3 - x^2}{x + 2} = q$, where q is a constant, has one repeated real root.
Find two possible values for q. (4 marks)

Selected exam question bank questions

Question 38

The expression $16 - 6x - x^2$ can be written in the form $q - (x + p)^2$, where p and q are integers.

a. Find the value of p and the value of q. (3 marks)

b. Calculate the discriminant of $16 - 6x - x^2$. (1 mark)

c. Sketch the curve with equation $y = 16 - 6x - x^2$, showing clearly the coordinates of any points where the curve crosses the coordinate axes. (3 marks)

Question 43

$f(x) = x^2 - 6x + 16$, $x \in \mathbb{R}$.

a. Show that $f(x)$ can be written in the form $(x - p)^2 + q$, where p and q are constants to be found. (2 marks)

b. Hence sketch the graph of $y = f(x)$ showing the coordinates of the turning point. (3 marks)

c. Given that the equation $f(x) = k$, where k is a constant, has two distinct real solutions, find the range of possible values of k. (3 marks)

Question 52

$f(x) = x^2 - 6x + 13$, $x \in \mathbb{R}$.

a. Express $f(x)$ in the form $(x - a)^2 + b$ where a and b are constants. (2 marks)

The curve C with equation $y = f(x)$ crosses the y-axis at the point P and has a minimum point at the point Q.

b. Sketch the graph of C, showing the coordinates of points P and Q. (3 marks)

c. Find the length of the line segment PQ, writing your answer as a simplified surd. (3 marks)

Question 67

A company makes a particular type of portable DVD player. The annual profit made by the company is modelled by the equation $P = 6.25(30x - x^2) - 1256.25$, where P is the profit measured in thousands of pounds and x is the selling price of the DVD player, in pounds.

a. **i** Write the model in the form $P = c - a(x - b)^2$, where a, b and c are constants to be found.
ii Hence, or otherwise, find the values of x which give $P = 0$. Give your answer to 3 significant figures.
iii Sketch the graph of P against x, clearly labelling the intersections with the x-axis and the maximum point on the graph. (7 marks)

b. Using the model, explain why £25 is not a sensible selling price for the DVD player. (2 marks)

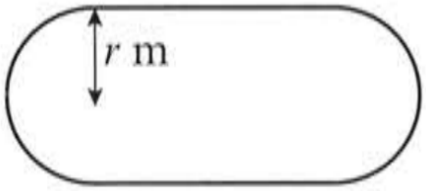
The company wishes to maximise its annual profit.

State, according to the model:

c. **i** the maximum possible annual profit
ii the selling price of the disc player that maximises the annual profit. (2 marks)

Question 91

A running track can be modelled by two straight lines, which are opposite sides of a rectangle, joined by two semicircular arcs each of radius r metres, as shown in the diagram.



The distance around the entire track is 500 m, and the track itself can be assumed to be a thin line.

a. The area inside the track is A m^2. Show that $A = 500r - \pi r^2$. (5 marks)

b. Hence find, in terms of π, the maximum value of A and the value of r at which this maximum occurs. (You do not have to justify that the value is a maximum.) (6 marks)

Answers

2.1 Solving quadratic equations

1. **a** $x = -5$ or $x = -2$ **b** $x = 8$ or $x = -3$ **c** $x = 0$ or $x = -6$

2. **a** $x = 0$ or $x = 3$ **b** $x = 3.5$ or $x = -3.5$ **c** $x = 2.5$ **d** $x = 2$ or $x = 0.5$

3. **a** $x = 4$ **b** $x = 3.5$ or $x = -3$ **c** $x = -3$ or $x = -4$

4. **a** $x = -4 \pm \sqrt{10}$ **b** $x = -2 \pm \sqrt{5}$ **c** $x = \frac{6 \pm \sqrt{6}}{2}$

5. **a** $x = 1$ or $x = 9$ **b** $x = \pm\frac{5}{4}$
c $x = \frac{5}{3}$ or $x = -5$ **d** $x = 3.53$ or $x = 0.472$
e $x = 0$ or $x = 5$ **f** $x = -0.177$ or $x = -2.82$

6.
a. $(2x + 3)(3x - 5) = 20 \Rightarrow 6x^2 - x - 15 = 20 \Rightarrow 6x^2 - x - 35 = 0$
b. Length = 8 m, width = 2.5 m

7. $-2 \pm \sqrt{3}$

8. $x = 0.485$ and $x = -16.5$

9. **a** 3.12 seconds **b** 1 and 2 seconds

2.2 Completing the square

1. **a** $(x + 1)^2 - 1$ **b** $(x - 4)^2 - 16$ **c** $(x - 6)^2 - 36$

2. **a** $(x - 4)^2 - 4$ **b** $(x - \frac{1}{2})^2 - \frac{49}{4}$ **c** $(x - \frac{3}{2})^2 - \frac{25}{4}$

3. **a** $3(x - 2)^2 + 5$ **b** $5(x - 1)^2 + 7$ **c** $3(x - \frac{7}{6})^2 - \frac{25}{12}$

4. **a** $x = 1 \pm \sqrt{11}$ **b** $x = \frac{-6 \pm \sqrt{6}}{5}$ **c** $x = \frac{7 \pm \sqrt{73}}{6}$

5. **a** $(x - 3.5)^2 - 14.25$ **b** -14.25

6. **a** $-3(x + \frac{1}{3})^2 + \frac{13}{3}$ **b** $x = \frac{-1 \pm \sqrt{13}}{3}$

7. **a** $7 - (x - 3)^2$ **b** $x = 3 \pm \sqrt{7}$

2.3 Functions

1. **a** 0 **b** 5 **c** 15.75 **d** 5

2. $x = 2$ and $x = -2$

3. **a** -2 and -7 **b** 9 and -9 **c** $\sqrt[3]{-3}$ and $\sqrt[3]{-4}$ **d** 0, -3 and 7

4. **a** $(x + 3)^2 - 7$ **b** $-3 \pm \sqrt{7}$ **c** -7, when $x = -3$

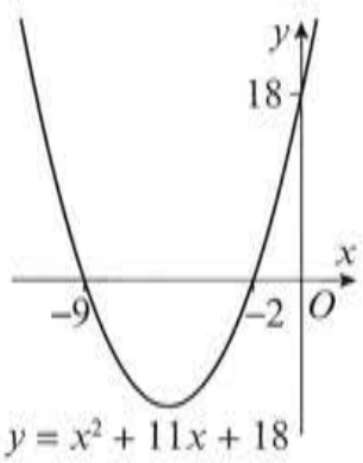
5.
a. $(x + 3)^2 + 4$
b. Smallest value of $(x + 3)^2$ is 0 so minimum value is 4.

6. **a** 1 and -1 **b** $\sqrt[5]{\frac{1}{3}}$ and $\sqrt[5]{\frac{1}{2}}$ **c** 1 and 4

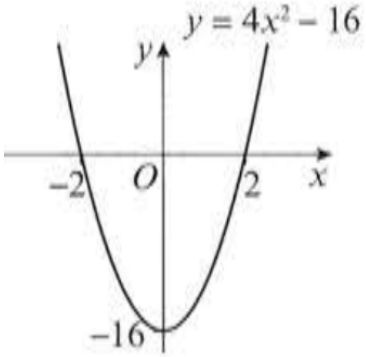
7. **a** $(2^x - 2)(2^x - 4)$ **b** 1 and 2

2.4 Quadratic graphs

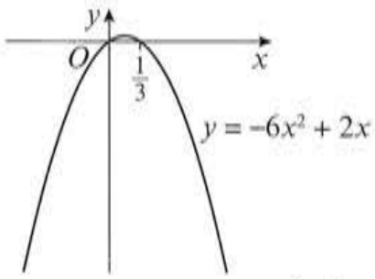
1. **a.**



b.

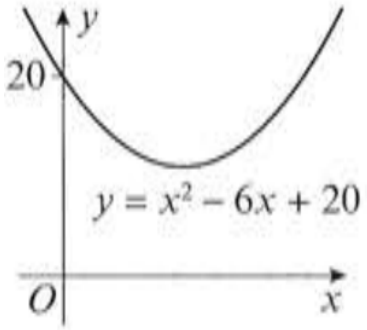


c.



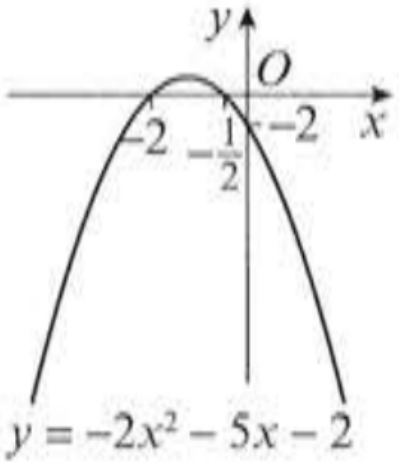
2. **a** $(1, 9)$ **b** $(-\frac{1}{2}, -\frac{25}{4})$ **c** $(-\frac{13}{2}, \frac{1}{4})$

3. **a.**



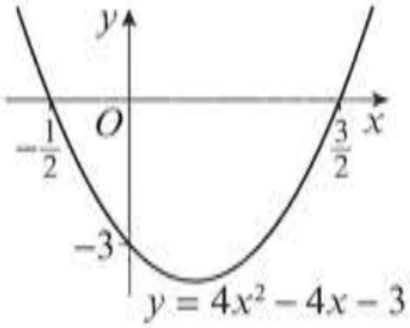
Turning point $(3, 11)$, line of symmetry $x = 3$

b.



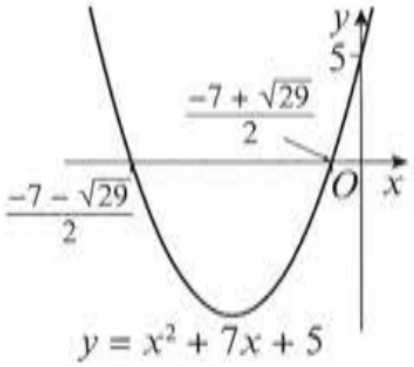
Turning point $(-\frac{5}{4}, \frac{9}{8})$, line of symmetry $x = -\frac{5}{4}$

c.



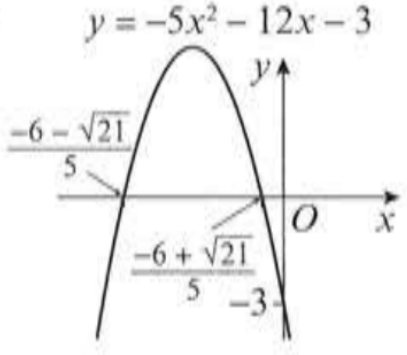
Turning point $(\frac{1}{2}, -4)$, line of symmetry $x = \frac{1}{2}$

4. **a.**



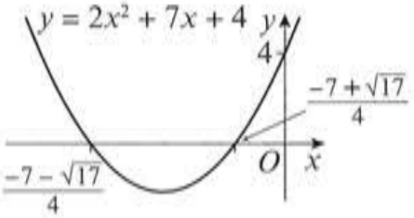
Turning point $(-\frac{7}{2}, -\frac{29}{4})$, line of symmetry $x = -\frac{7}{2}$

b.



Turning point $(-\frac{6}{5}, \frac{21}{5})$, line of symmetry $x = -\frac{6}{5}$

c.

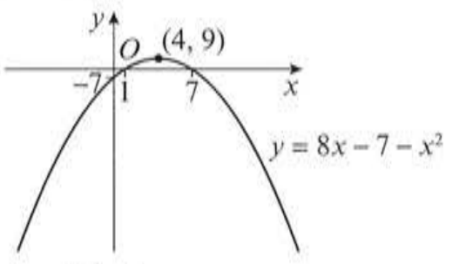


Turning point $(-\frac{7}{4}, -\frac{17}{8})$, line of symmetry $x = -\frac{7}{4}$

5.

a. $p = 4$, $q = 9$

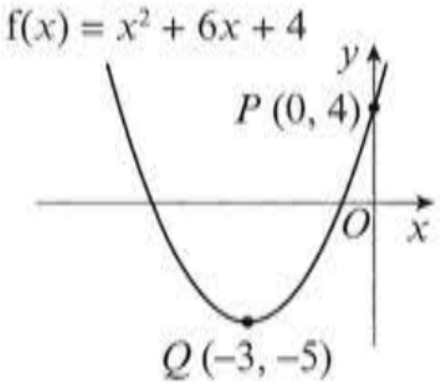
b.



6.

a. $(x+3)^2 - 5$

b.

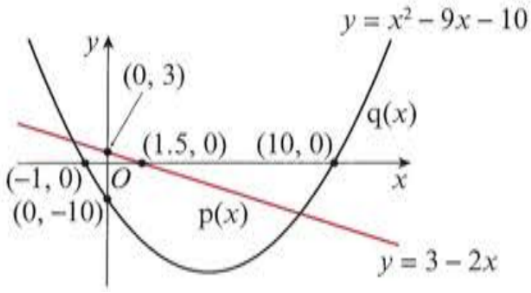


c. The smallest y-value of the curve in part **b** is -5, so $f(x) = -6$ cannot have real solutions.

7.

a. $x = 10$ or $x = -1$

b.



8. $a = 5$, $b = -20$, $c = 15$

2.5 The discriminant

1. **a** $b^2 - 4ac > 0$ **b** $b^2 - 4ac = 0$ **c** $b^2 - 4ac < 0$

2. **a** **i** 25 **ii** 2 roots
b **i** 0 **ii** 1 root
c **i** -72 **ii** no roots

3. **a** 16, ii **b** 0, iii **c** -12, i

4. **a** $k^2 - 100$ **b** $k = \pm 10$

5. **a** $k^2 + 2k + 9$ **b** $k^2 + 2k + 9 = (k+1)^2 + 8$
c $(k+1)^2 + 8 \geqslant 8 > 0$

6. $k > -\frac{9}{20}$

7. $p > \frac{49}{8}$

2.6 Modelling with quadratics

1. **a** 25 m **b** $-0.04(x-12.5)^2+6.25$ **c** 6.25 m, 12.5 m

2.
- **a.** $P=73.5x-5.25x^2-107.5 \Rightarrow P=-5.25(x^2-14x)-107.5 \Rightarrow P=149.75-5.25(x-7)^2$
- **b.** **i** £149 750 **ii** £7

3.
- **a.** **i** $(1.25,-3)$ **ii** $y=1.92(x-1.25)^2-3$
- **b.** For a depth of 1 m, the boat would need to be less than 2.04 m wide. Hence boat cannot fit through.

4. **a** 0 workers produce 0 cars
b 420 **c** 500 cars, 2500 workers

5.
- **a.** The ball was thrown from a height of 120 m.
- **b.** $\frac{4085}{32}-4.9(t-\frac{5}{4})^2$
- **c.** **i** 6.35 seconds (3 s.f.)
  ii 128 metres (3 s.f.), 1.25 seconds

6.
- **a.** Negative profit can be interpreted as a loss, so the model may still be valid.
- **b.** $P=160-6.25(x-10)^2$
- **c.** £7.48
- **d.** **i** £160 000 **ii** £10

7. 2000, £24

Problem solving: Set A

Bronze **a** $1-(x-1)^2$ **b** 2 m

Silver

a The stone was thrown from a height of 114 m

b i $10.4x-5.2x^2=-5.2(x^2-2x) \Rightarrow h(x)=114-5.2(x^2-2x)$

ii $h(x)=119.2-5.2(x-1)^2$

c i 5.79 m (3 s.f.) **ii** 119.2 m, 1 m

Gold **a** The stone was thrown from a height of 125 m

b 134 m (3 s.f.) **c** 125 m (3 s.f.)

Problem solving: Set B

Bronze **a** $4^2-8k^2=0 \Rightarrow 16-8k^2=0$ **b** $\pm\sqrt{2}$

Silver 24

Gold -6 and 2

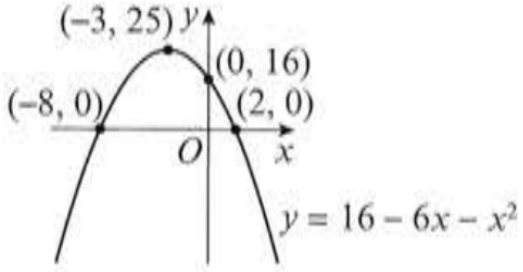
Selected exam question bank answers

Question 38

a. $p = 3$, $q = 25$

b. 100

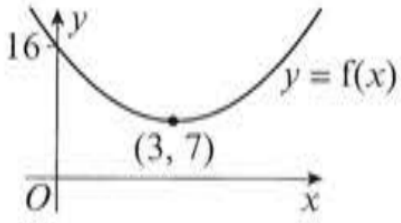
c.



Question 43

a. $f(x) = (x-3)^2 - 9 + 16 = (x-3)^2 + 7$

b.

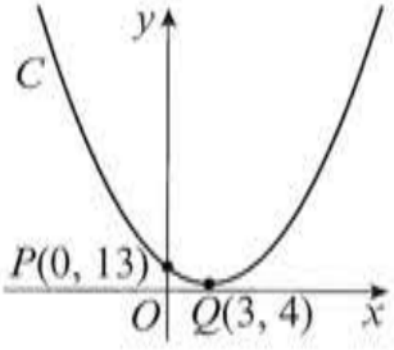


c. $k > 7$

Question 52

a. $(x-3)^2 + 4$

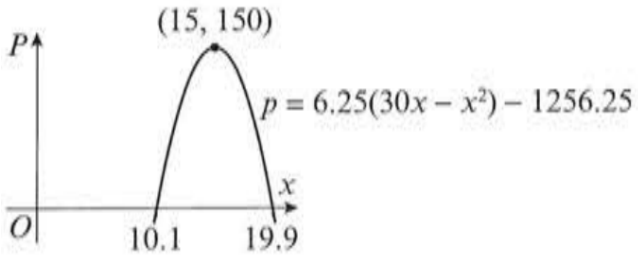
b.



c. $3\sqrt{10}$

Question 67

a. **i** $P = 150 - 6.25(x - 15)^2$ **ii** 10.1 and 19.9
iii



b. The company would make an annual loss of £475 000.

c. **i** £150 000 **ii** £15

Question 91

a. Perimeter $= 500 = 2\pi r + 2x \Rightarrow x = 250 - \pi r$
Area $= \pi r^2 + 2rx = \pi r^2 + 2r(250 - \pi r) = 500r - \pi r^2$

b. $A = \frac{62\,500}{\pi}$, $r = \frac{250}{\pi}$